

Validity

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Validity

- Valid research is that which accurately reflects the real world phenomena it is intended to describe (and sometimes, to explain) (deVellis 2011)
- Measurement validity: validly measure what we intent to measure
- Internal validity refers to researchers' claims that the data they have measured reveal a causal process at work.
- External validity refers to the representativeness, or generalizability of research findings

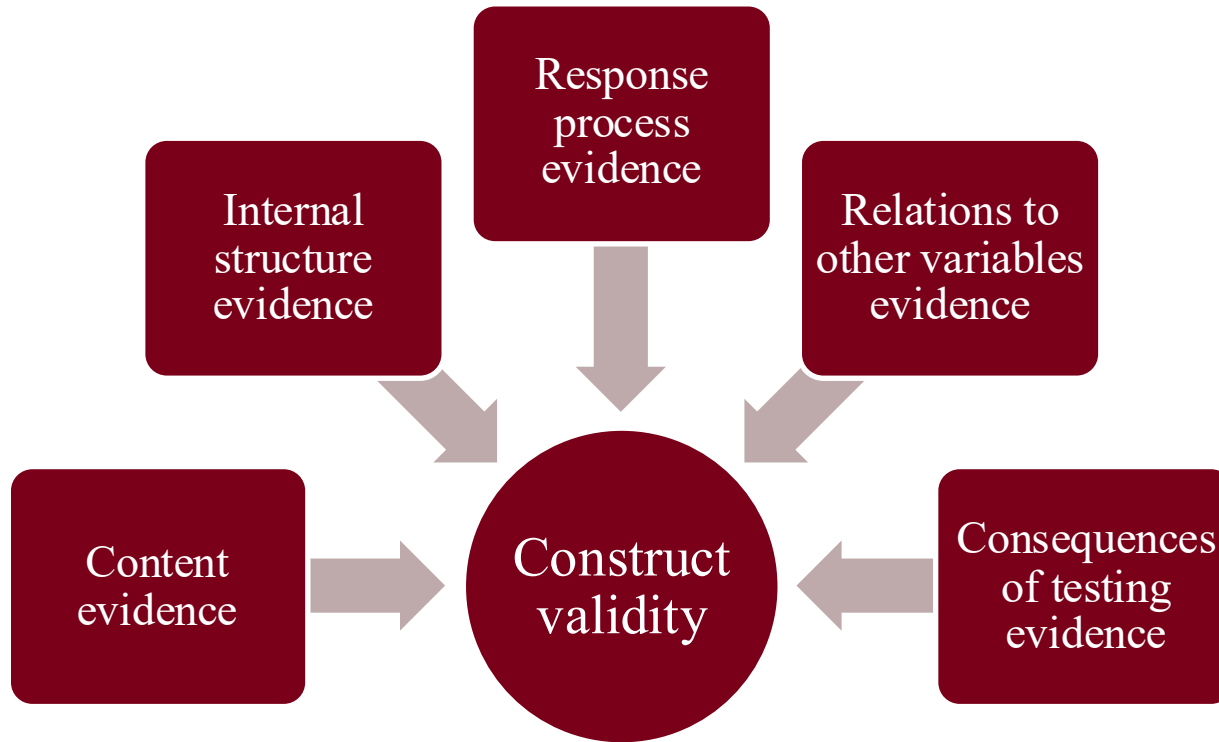


Definition of measurement validity

- 1. Validity is about the interpretation and the use of test scores in terms of a specific psychological construct; it is not about the test itself.
- 2. Validity is a matter of degree; it is not an “all-or-none” issue.
- 3. Validity is based on solid empirical evidence and theory.



Contemporary facets of validity



Contemporary facets of validity

Facet of Validity	Definition: The Match Between . . .	Typical Relevant Evidence
Content	The <i>actual</i> content of a test and the content that <i>should be</i> included in the test	Expert ratings of test content
Internal Structure	The <i>actual</i> internal structure of a test and the structure that the test <i>should</i> possess	Factor analysis of responses to test items (exploratory and/or confirmatory factor analysis)
Response Process	The psychological processes that respondents <i>actually</i> use when completing a measure and the processes that they <i>should</i> use	Interviewing respondents about their response processes. Also, indirect indicators such as response times, eye tracking
Associations	A measure's <i>actual</i> associations with other measures and the associations that the test <i>should have</i> with the other measures	Correlations with other measures, including relevant group variables. Convergent, discriminant, concurrent, predictive correlations
Consequences	The <i>actual</i> consequences of using a measure and the consequences that <i>should be</i> seen	Evaluation of intended consequences, differential impact, and systemic changes

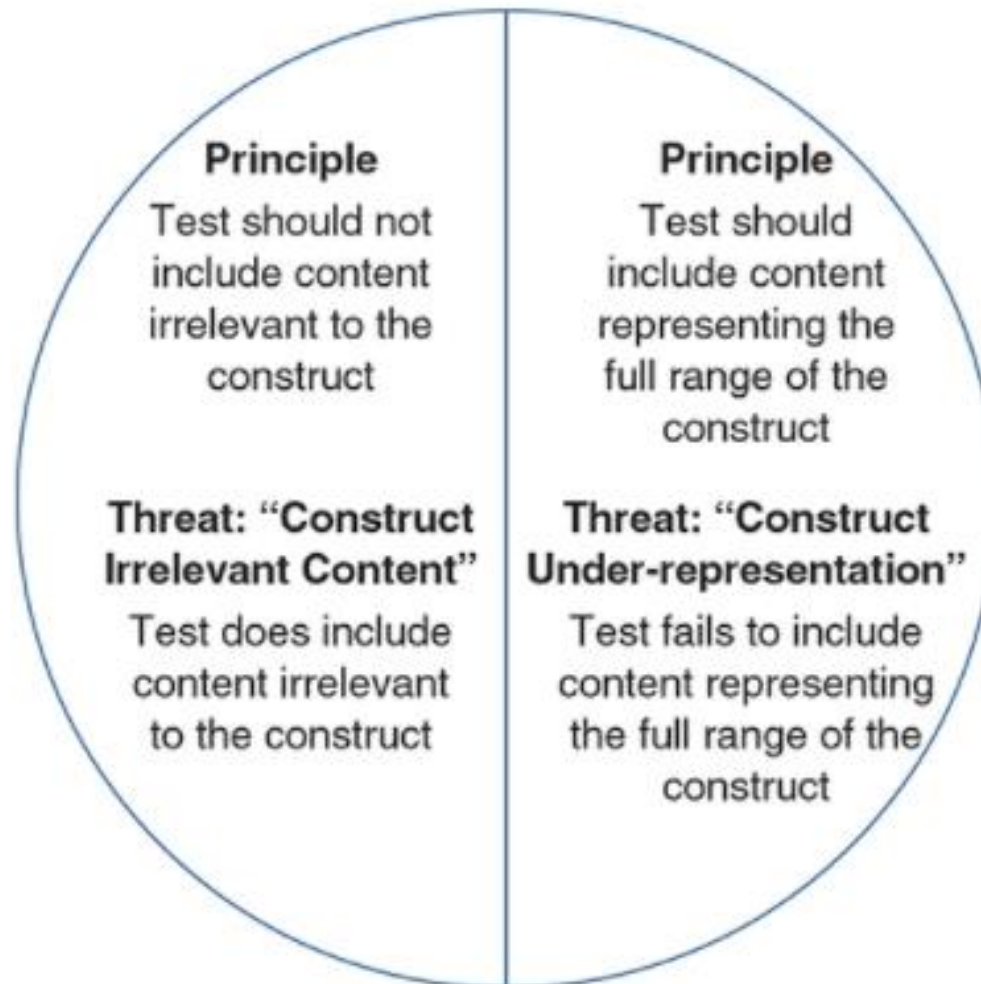


Evidence Based on Test Content

- The degree to which elements of an assessment instrument are relevant to and representative of the targeted construct for a particular assessment purpose.



Threats to Content Validity



Threats to Content Validity: Construct underrepresentation

- Construct underrepresentation refers to a situation in which test content is defined too narrowly, leaving out important aspects of the construct.
- Construct underrepresentation can also occur if a test covers the intended content adequately but does not do so at the intended cognitive level.



Threats to Content Validity: Construct-irrelevant variance

- Construct-irrelevant variance occurs when test scores are influenced by factors that are not part of the intended construct.
 - construct-irrelevant easiness
 - construct-irrelevant difficulty



Content validity v.s. Face validity

- Face validity answers the question, “Does it appear to the test taker that the questions on the test are related to the purpose for which the test is being given?”
- Face validity is never an acceptable means of demonstrating evidence of the validity of a test
- Face validity might have important implications for the test’s use



Evidence Based on Internal Structure

- Evidence based on internal structure concerns the internal structure or dimensionality of the test
- the match between the actual internal structure of a test and the structure that the test should possess
 - Rosenberg's (1989) Self-Esteem Inventory was designed to be unidimensional
 - The Multidimensional Self-Esteem Inventory (MSEI; O'Brien & Epstein, 1988) was designed to measure global self-esteem along with eight components of self-esteem.



Evidence Based on Internal Structure

- Exploratory factor analysis
- Confirmatory factor analysis

Number of Factors	Associations Among Factors	Meaning of Each Factor
Explanation The actual number of factors within a test should match the number of dimensions within the construct(s).	Explanation The actual associations among the factors within a test should match the associations among the dimensions of the construct(s).	Explanation The meaning of the actual factors within a test (as revealed by factor loadings) should match the meaning of the dimensions of the construct(s).
Example Global self-esteem is a unidimensional construct, so the RSEI should have only one factor.	Example There are sub-components of self-esteem that are connected to each other, so the inter-factor associations within the MSEI should match those	Example There are specific sub-components of self-esteem, so the factors within the MSEI should match those components.

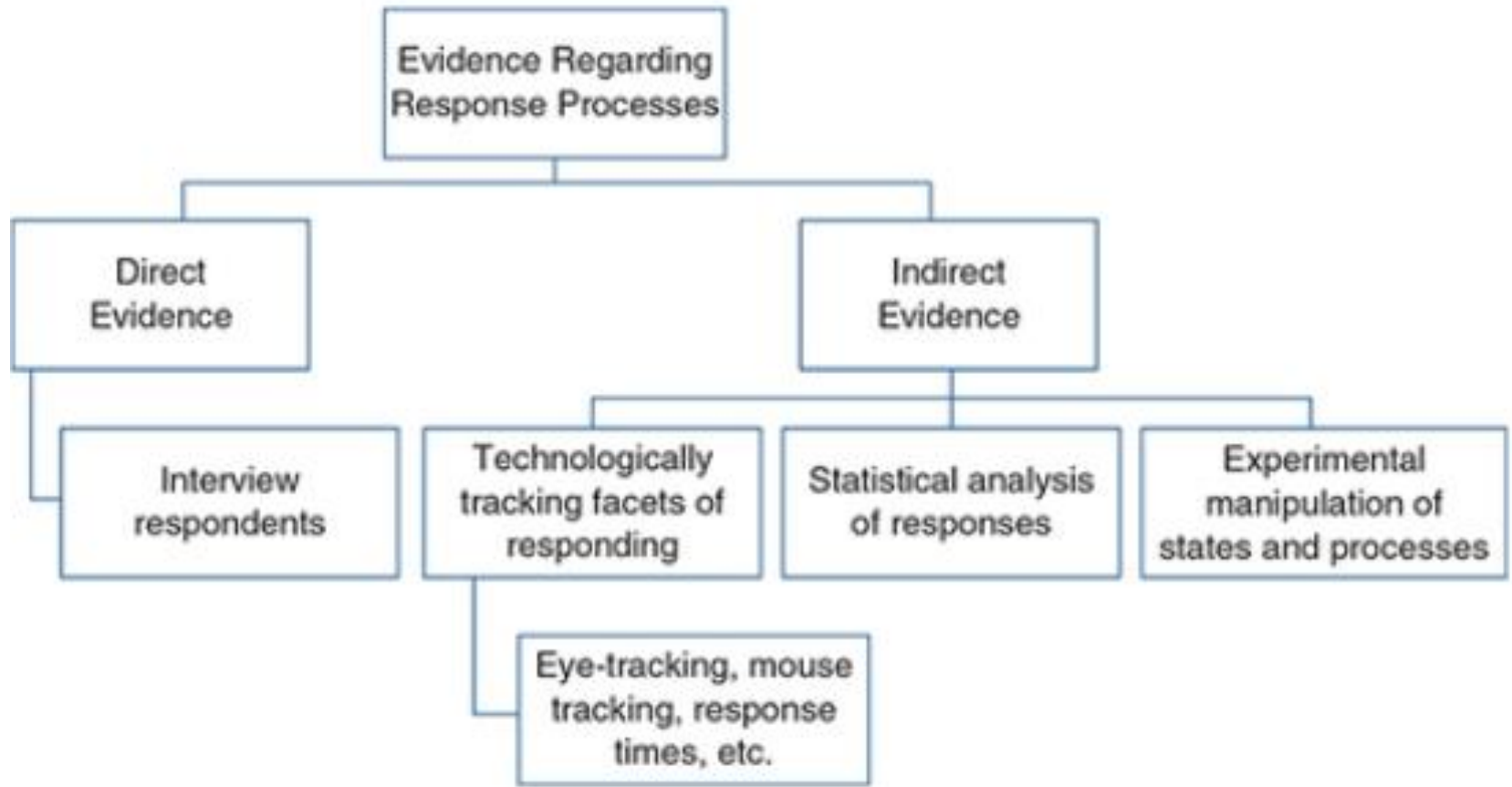


Evidence Based on Response Processes

- Evidence based on response processes concerns the match between the psychological processes that respondents actually use when completing a measure and the processes that they should use.
- Score interpretations are based on the assumption that certain cognitive processes have been used



Evidence Based on Response Processes



Evidence Based on Association with Other Variables

- Evidence based on association with other variables involves the match between a measure's actual associations with other measures and the associations that the test should have with those other measures.
- If a test's actual correlations with the other measures do indeed match the correlations suggested by the nature of a particular construct, then we gain support for the interpretation of the test as a measure of that construct.



Different types of validity evidence based on association with other variables

Convergent validity	Test scores are actually correlated (either positive or negatively) with other measures that they <i>should be</i> correlated with.	Are RSEI scores positively correlated with a measure of happiness and negatively correlated with a measure of depression, consistent with our understanding of global self-esteem?
Criterion validity	Test scores are actually related to a particularly important criterion variable that they should be correlated with.	Are SAT scores correlated with GPA?
Concurrent validity	Test scores are actually related to a particularly important criterion variable that is measured at the same time that the test itself is completed.	Are SAT scores (taken while in high school) correlated with high school GPA?
Predictive validity	Test scores are actually related to a particularly important criterion variable that is measured at a later time (than when the test itself is completed).	Are SAT scores (taken while in high school) correlated with college GPA?
Discriminant validity	Test scores are actually uncorrelated (or weakly correlated) with other measures that they <i>should not be</i> correlated with.	Are RSEI scores uncorrelated (or weakly correlated) with a measure of intelligence, consistent with our view that global self-esteem is unrelated to intelligence?



Evidence based on relations to other variables

- Test–criterion relationships
- Convergent evidence
- Discriminant evidence



Evidence based on relations to other variables

- Test–criterion relationships refer to situations in which test scores are used to predict future performance or current status on some criterion:
 - Predictive validity
 - the extent to which a score on a scale or test predicts scores on some criterion measure
 - Example: GRE
 - Concurrent validity
 - The extent to which one test can be used as an alternative to another
 - Example: shorter form of a longer test; computerized version vs paper-and-pencil test



Evidence based on relations to other variables

- Convergent evidence
 - test scores should be related to scores from other tests of the same or similar constructs
- Discriminant evidence
 - test scores should be less strongly related to scores from tests measuring dissimilar constructs



Internal Validity

- Internal validity refers to how well a researcher's survey design is testing the true relationship between the independent variable and dependent variable (Chambliss and Schutt 2013).
- Internal validity is established when a researcher has provided sound arguments, adequate statistical controls, and rigorous research design to establish a causal relationship between two correlated variables.

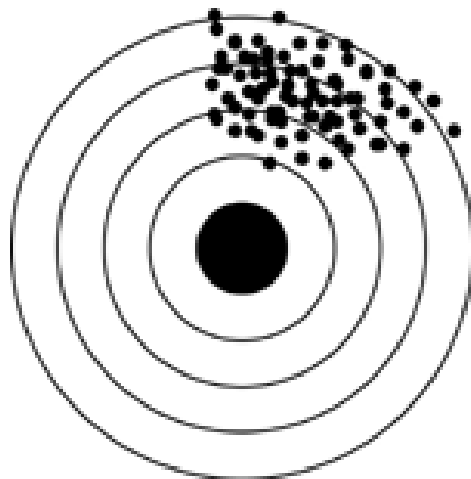


External Validity

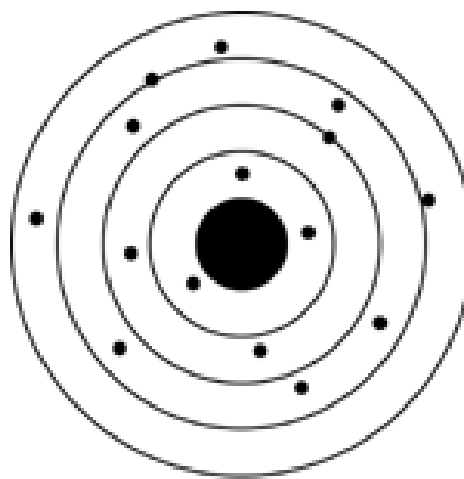
- External validity is the extent to which research findings can be generalized to other populations, environments, times, and settings (Flick 2011).



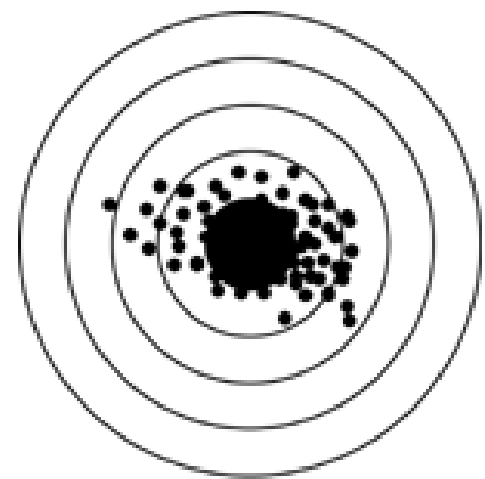
Reliability and measurement validity



Reliable but Not Valid



Valid but Not Reliable



Valid and Reliable